OPENNEBULA INSTALLATION

PROMETHEUS PREPARATION

Before we start working on our computers we need to prepare our Prometheus VM. Login to your allocated computer using your allocated VM username and localhost1 as the password. Open a terminal (right click anywhere on the desktop) and change your password using the *passwd* command. Once that is done, go to your Home folder and then go to the Global folder. In there you will find a TB2-Prometheus.tar file. Extract it into your home folder and a folder named vmware containing TB2-Prometheus should be created. Your VM is now in place.

Go to the top left corner to *Applications – System Tools – Vmware Player* (you can drag and drop this on the Desktop for easy access in the future). VMPlayer will start, so once you sort out the license agreement etc you will be presented with the Player's GUI. Choose **Open a Virtual Machine** from the options on the right and select **the TB2-Prometheus** folder and then choose the **TB2-Prometheus.vmx** file. Prometheus will now load in the Player ready to run so click **Power On**. Select the *"I copied it"* option (if such a message shows up). At the Login Screen click on the little grey link under the username that says ***Not Listed*** and in the box that appears type **root** as the username, press Enter and then **localhost1** as the password. You must ***ALWAYS LOGIN AS ROOT!*** Once you are logged in Set the Virtual Machine to Full Screen (**Alt-CTRL-Enter**). You never need to do anything else on the Host.

Now you need to sort out a few details here, namely Networking. The VM at this moment has an ip address of 148.197.28.253 You will need to change that to correspond to your computer's IP address. Look at the sticker on the bottom left corner of your monitor and take a note of your VLAN number. This is the IP address of your host. Go back to your Prometheus VM, go to the top right corner on the little computer icon next to the little speaker and click it. Select the Wired Connected Network card from the drop-down menu that appears and then **Wired Settings** under that. Click on the little **cog** for settings next to the ON/OFF button on the right. Click on **IPV4**. Change the IP address to: **148.197.28."host+50"**. Click **Apply** on the top right corner. Turn the network card OFF and then ON by toggling the switch next to the cog. Close the network settings window, right click anywhere on the desktop and select Open Terminal. Try the command: *ping -c3 google.com*. You should get 3 replies. Your network is configured.

You can now use Google Chrome in Prometheus to go to Moodle and get the labsheets and any other files you need for these sessions. Please note we use Chrome for compatibility with google docs where your logbook will be. You can use Firefox if you wish but you might have issues pasting screenshots in your logbook if you use it.

FRONT-END INSTALLATION

This guide shows you how to install OpenNebula from the binary packages. Using the packages provided in the OpenNebula site is the recommended method, to ensure the installation of the latest version and to avoid possible packages divergences of different distributions. There are two alternatives here: you can add **our package repositories** to your system or visit the software menu to **download the latest package** for your Linux distribution. We're going with the first option.

ADD OPENNEBULA REPOSITORIES

To add the OpenNebula repository create the following file as root:

gedit /etc/yum.repos.d/opennebula.repo

```
[opennebula]
name=OpenNebula Community Edition
baseurl=https://downloads.opennebula.io/repo/6.2/CentOS/7/$basearch
enabled=1
gpgkey=https://downloads.opennebula.io/repo/repo.key
gpgcheck=1
repo_gpgcheck=1
```

INSTALLING THE SOFTWARE

Before installing:

Activate the EPEL repo. In CentOS this can be done with the following 2 commands:

```
yum makecache
yum install epel-release
```

There are packages for the Front-end, distributed in the various components that conform OpenNebula, and packages for the virtualization host. To install a CentOS/RHEL OpenNebula Front-end with packages from **our repository**, execute the following as root.

```
yum install opennebula-server opennebula-sunstone
yum install opennebula-gate opennebula-flow
```

STARTING OPENNEBULA

Log in as the `oneadmin` user to follow these steps:

su oneadmin

```
echo "oneadmin:localhost1" > ~/.one/one_auth
```

exit

Finally before we start the OpenNebula services it's a good idea to disable the new FireEdge GUI as we will manage our Cloud through the web interface (Sunstone) .

gedit /etc/one/sunstone-server.conf

and at the very end of this file find the last two lines and add a # in front of them like so:

#:private_fireedge_endpoint: http://localhost:2616
#:public_fireedge_endpoint: http://localhost:2616

You are ready to start the OpenNebula daemons. You can use systemctl for Linux distributions which have adopted system (switch to root user again):

```
systemctl start opennebula
systemctl start opennebula-sunstone
```

VERIFYING THE INSTALLATION

After OpenNebula is started for the first time, you should check that the commands can connect to the OpenNebula daemon. You can do this in the Linux CLI or in the graphical user interface: Sunstone.

LINUX CLI

In the Front-end, run the following command as oneadmin:

```
oneuser show

USER 0 INFORMATION
ID              : 0
NAME            : oneadmin
GROUP           : oneadmin
PASSWORD        : 3bc15c8aae3e4124dd409035f32ea2fd6835efc9
AUTH_DRIVER     : core
ENABLED         : Yes
```

```
USER TEMPLATE
TOKEN_PASSWORD="ec21d27e2fe4f9ed08a396cbd47b08b8e0a4ca3c"

RESOURCE USAGE & QUOTAS
```

The OpenNebula logs are located in `/var/log/one`, you should have at least the files `oned.log` and `sched.log`, the core and scheduler logs. Check `oned.log` for any error messages, marked with `[E]`.

SUNSTONE

Now you can try to log in into Sunstone web interface. To do this point your browser to

`http://localhost:9869`.

If everything is OK you will be greeted with a login page. The user is `oneadmin` and the password is the one in the file `/var/lib/one/.one/one_auth` (so localhost1) in your Front-end.

DIRECTORY STRUCTURE

The following table lists some notable paths that are available in your Front-end after the installation:

Path	Description
`/etc/one/`	Configuration Files
`/var/log/one/`	Log files, notably: `oned.log`, `sched.log`, `sunstone.log` and `<vmid>.log`
`/var/lib/one/`	`oneadmin` home directory
`/var/lib/one/datastores/<dsid>/`	Storage for the datastores

Path	Description
`/var/lib/one/vms/<vmid>/`	Action files for VMs (deployment file, transfer manager scripts, etc...)
`/var/lib/one/.one/one_auth`	`oneadmin` credentials
`/var/lib/one/remotes/`	Probes and scripts that will be synced to the Hosts
`/var/lib/one/remotes/hooks/`	Hook scripts
`/var/lib/one/remotes/vmm/`	Virtual Machine Manager Driver scripts
`/var/lib/one/remotes/auth/`	Authentication Driver scripts
`/var/lib/one/remotes/im/`	Information Manager (monitoring) Driver scripts
`/var/lib/one/remotes/market/`	MarketPlace Driver scripts
`/var/lib/one/remotes/datastore/`	Datastore Driver scripts
`/var/lib/one/remotes/vnm/`	Networking Driver scripts
`/var/lib/one/remotes/tm/`	Transfer Manager Driver scripts

KVM NODE INSTALLATION

Execute the following commands to install the node package and restart libvirt to use the OpenNebula provided configuration file. Close your web browser sunstone tab first!

```
yum install opennebula-node-kvm.noarch
service libvirtd restart
```

CONFIGURE PASSWORDLESS SSH

OpenNebula will need to SSH passwordlessly from any node (including the frontend) to any other node. Add the following snippet to ~/.ssh/config as oneadmin so it doesn't prompt to add the keys to the known_hosts file:

```
su - oneadmin
```

```
vi ~/.ssh/config

once the file is open press INSERT on your keyboard to go into edit mode and navigate towards the end of the file at line 37:

Host *

    StrictHostKeyChecking no

    UserKnownHostsFile /dev/null

(save the file and exit back to the terminal by pressing ESC then typing :wq)

chmod 600 ~/.ssh/config
```

You should verify that connecting from the Front-end, as user `oneadmin`, to the nodes and the Front-end itself, and from the nodes to the Front-end, does not ask password:

```
ssh localhost
exit
```

TRY OUT SUNSTONE!

Go to http://localhost:9869

Login: oneadmin

Password: (whatever is in your ~/.one/one_auth so hopefully localhost1)

What sort of information can you see in the Dashboard area?